Supplementary Material: Issues with Measuring Task Complexity via Random Policies in Robotic Tasks

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1 POSITIONAL ERRORS AT END-EFFECTOR

This section demonstrates how joint errors propagate to the end effector. Given a planar serial chain with n joints, the relationship between the end-effector velocities $\dot{x} \in \mathbb{R}^2$ and joint velocities $\dot{\theta} \in \mathbb{R}^n$ is (Sciavicco and Siciliano, 2012):

$$\dot{x} = J(\theta)\dot{\theta} \tag{1}$$

where $J(\theta) \in \mathbb{R}^{2\times n}$ is the Jacobian matrix. The $k-$th column of $J(\theta)$ is given by:

$$\begin{aligned} J_k(\theta) &= \frac{\partial x}{\partial \theta_k} \\ &= \begin{bmatrix} -\sum_{i=k}^{n} l_i \sin\left(\sum_{j=1}^{i} \theta_j\right) \\ \sum_{i=k}^{n} l_i \cos\left(\sum_{j=1}^{i} \theta_j\right) \end{bmatrix} \end{aligned} \tag{2}$$

whose Euclidean norm is

$$\begin{aligned} \|J_k(\theta)\|_2 &= \sqrt{\left(\sum_{i=k}^{n} l_i \sin\left(\sum_{j=1}^{i}\theta_j\right)\right)^2 + \left(\sum_{i=k}^{n} l_i \cos\left(\sum_{j=1}^{i}\theta_j\right)\right)^2} \\ &= \sqrt{\sum_{i=k}^{n} l_i^2 + 2\sum_{k\leq i<h\leq n} l_i l_h \cos\left(\sum_{j=1}^{i}\theta_j - \sum_{j=1}^{h}\theta_j\right)} \\ &\leq \sum_{i=k}^{n} l_i \end{aligned} \tag{3}$$

Given a small joint error, we can use Equation 1 to estimate the error at the end-effector as:

$$\begin{aligned} \delta x &\approx J(\theta)\delta\theta \\ &= \sum_{k=1}^{n} J_k \delta\theta_k \end{aligned} \tag{4}$$

Therefore, the magnitude of the $||\delta x||_2$ can be bounded using the triangular inequality as follows:

$$\begin{aligned} \|\delta x\|_2 &= \left\|\sum_{k=1}^{n} J_k \delta\theta_k\right\|_2 \\ &\leq \sum_{k=1}^{n} \|J_k \delta\theta_k\|_2 \\ &= \sum_{k=1}^{n} \|J_k\|_2 \, |\delta\theta_k| \end{aligned} \tag{5}$$

If $|\delta\theta_k| \leq \epsilon$ for $k = 1, \cdots, n$, and we substitute Equation 3 into Equation 5, then

$$
\begin{aligned}
\|\delta x\|_2 &\leq \sum_{k=1}^{n} \|J_k\|_2 \, \epsilon \\
&\leq \epsilon \sum_{k=1}^{n} \sum_{i=k}^{n} l_i \\
&= \epsilon \sum_{i=1}^{n} \sum_{k=1}^{i} l_i \\
&= \epsilon \sum_{i=1}^{n} i l_i
\end{aligned} \tag{6}
$$

The final results of Equation 6 demonstrate that the error magnitude at the end-effector is exacerbated by both the number of joints n and link lengths l_i. Therefore, the task complexity should increase when degree of freedom n increase and/or when the link lengths l_i increases, since an RL agent has to deal with increased error sensitivity.

2 SAC ARCHITECTURE

In this section, we specify the architecture of the SAC algorithm used to produce the learning performances in Figure 2. Table 1 entails a list of the configuration parameters for the algorithm. The ADAM (Kingma and Ba, 2017) optimiser was used in both the actor and critic (neural-network) models.

Table 1: SAC Hyperparameters.

Parameter	Value
learning rate	0.001
discount factor (γ)	0.99
temperature coefficient (α)	0.2
number of hidden layers (all networks)	2
number of hidden units per layer	256
number of samples per minibatch	64
nonlinearity	ReLU
target smoothing coefficient (τ)	0.005
Experience Replay size	10^6

3 ESTIMATION METHODS

3.1 PIC and POIC values

This section describes how the mean and standard deviation of the PIC and POIC values presented in Table 1 were determined. We used bootstrap resampling of the 10^4 samples to estimate the distribution of PIC and POIC. This allowed us in addition to compute the confidence intervals shown in Table 2.

3.2 Performance Distribution Plots

This section presents performance distribution plots (in Figure 1) for the tasks *1-link (L=1.0m)* and *1-link (L=1.65m)* arms with sparse rewards.

3.3 Statistical Significance Testing

We depict results for using Welch's t-test (West, 2021) for significance assessment. Table 3 presents the results. The negative statistic denotes the incorrect order of task hardness according to PIC/POIC. We observe that the p-values are less than 0.005, which makes the differences of PIC and POIC values across tasks statistically significant.

3.4 Initialisation Methods

In RWG, we mainly used multivariate normal distribution prior $p(\theta) = \mathcal{N}(0, I)$, where $I \in \mathbb{R}^{d \times d}$ given weight vectors $\theta_n \in \mathbb{R}^d$, to sample weights $\theta_n \sim p(\theta)$. This was combined with a neural network architecture (NN) of 2 hidden layers (HL), each with 32 units (HU) and the NN is without bias, i.e. NN = [2HL, 32HU, w/o bias]. In Figure 2, we show that performance distributions for the tasks are unchanging when prior distribution is altered. We tested with the following cases:

Table 2: Complete PIC and POIC values based on bootstrapping.

Rewards	Tasks	Metric	Value	CI (95%)
Dense	*1-link* ($L = 1m$)	PIC	4.005 ± 0.0085	$(3.979; 4.012)$
Dense	*1-link* ($L = 1m$)	POIC	$(2.628 \pm 0.056) \cdot 10^{-3}$	$(2.520; 2.738) \cdot 10^{-3}$
Dense	*1-link* ($L = 1.65m$)	PIC	4.153 ± 0.0084	$(4.126; 4.159)$
Dense	*1-link* ($L = 1.65m$)	POIC	$(4.105 \pm 0.085) \cdot 10^{-3}$	$(3.944; 4.279) \cdot 10^{-3}$
Dense	*2-link*	PIC	4.205 ± 0.0061	$(4.195; 4.219)$
Dense	*2-link*	POIC	$(0.722 \pm 0.011) \cdot 10^{-3}$	$(0.701; 0.746) \cdot 10^{-3}$
Sparse	*1-link* ($L = 1m$)	PIC	0.0851 ± 0.0004	$(0.0843; 0.0859)$
Sparse	*1-link* ($L = 1m$)	POIC	$(1.982 \pm 0.034) \cdot 10^{-3}$	$(1.911; 2.047) \cdot 10^{-3}$
Sparse	*1-link* ($L = 1.65m$)	PIC	0.0713 ± 0.0002	$(0.0708; 0.0718)$
Sparse	*1-link* ($L = 1.65m$)	POIC	$(1.197 \pm 0.031) \cdot 10^{-3}$	$(1.106; 1.233) \cdot 10^{-3}$
Sparse	*2-link*	PIC	$0.0460 \pm 6.37 \cdot 10^{-5}$	$(0.0458; 0.0461)$
Sparse	*2-link*	POIC	$(0.946 \pm 0.0079) \cdot 10^{-3}$	$(0.932; 0.962) \cdot 10^{-3}$

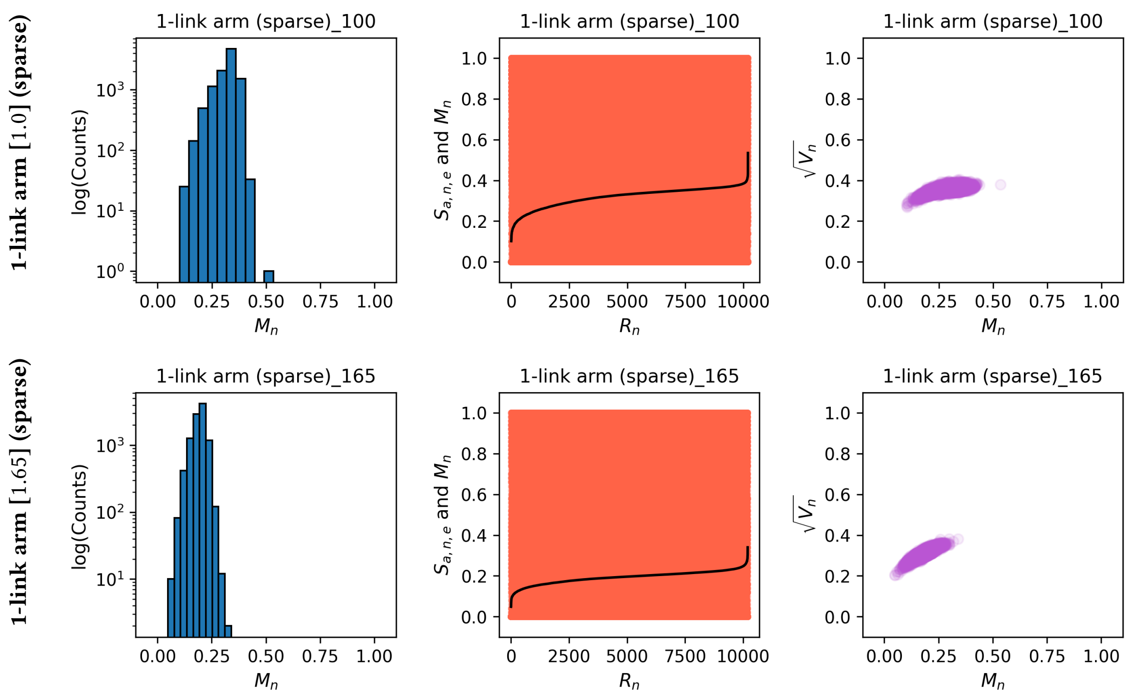

Figure 1: Performance distribution plots for the tasks *1-link (L=1.0m)* and *1-link (L=1.65m)* arms with sparse rewards. The performance is normalised using min-max scaling to allow tasks to have the same range.

(1) Default: $p(\theta) = \mathcal{N}(0, I)$ + NN = [2HL, 32HU, w/o bias]
(2) w/ Bias: $p(\theta) = \mathcal{N}(0, I)$ + NN = [2HL, 32HU, w/ bias]
(3) Variance: $p(\theta) = \mathcal{N}(0, 2I)$ + NN = [2HL, 32HU, w/o bias]
(4) Units64: $p(\theta) = \mathcal{N}(0, I)$ + NN = [2HL, 64HU, w/o bias]
(5) Units256: $p(\theta) = \mathcal{N}(0, I)$ + NN = [2HL, 256HU, w/o bias]
(6) Uniform: $p(\theta) = Unif(-1, 1)$ + NN = [2HL, 32HU, w/o bias]

To attain the results in Figure 2, we used $N = 10^3$ samples (i.e. random policies) in each setting, where each sample was deploy across 500 episodes.

Table 3: Statistical significance between PIC and POIC values using Welch's t-test. Results are presented as: *statistic (p-value)*. D stands for dense-rewards and S stands for sparse-rewards.

	1-link [1.0](D)	*1-link [1.65](D)*	*2-link [0.95,1.7](D)*	*1-link [1.0](S)*	*1-link [1.65](S)*
			PIC		
1-link [1.0](D)	0.0 (1.0)	-	-	-	-
1-link [1.65](D)	-20.1 (2.3×10^{-7})	0.0 (1.0)	-	-	-
2-link [0.95,1.7](D)	-29.3 (2.9×10^{-8})	-8.6 (2.6×10^{-5})	0.0 (1.0)	-	-
1-link [1.0](S)	948 (4.2×10^{-12})	646 (2.8×10^{-11})	618 (3.4×10^{-11})	0.0 (1.0)	-
1-link [1.65](S)	955 (5.7×10^{-12})	649 (3.1×10^{-11})	621 (3.7×10^{-11})	25.4 (4.7×10^{-8})	0.0 (1.0)
2-link [0.95,1.7](S)	964 (6.9×10^{-12})	654 (3.3×10^{-11})	626 (3.9×10^{-11})	85.7 (6.5×10^{-8})	84.5 (3.3×10^{-8})
			POIC		
1-link [1.0](D)	0.0 (1.0)	-	-	-	-
1-link [1.65](D)	-17.7 (1.2×10^{-7})	0.0 (1.0)	-	-	-
2-link [0.95,1.7](D)	30.8 (4.5×10^{-6})	59.7 (2.6×10^{-7})	0.0 (1.0)	-	-
1-link [1.0](S)	8.1 (4.8×10^{-5})	27.9 (3.3×10^{-9})	-24.2 (3.4×10^{-5})	0.0 (1.0)	-
1-link [1.65](S)	22.3 (4.6×10^{-6})	49.0 (6.6×10^{-8})	-21.6 (1.5×10^{-6})	14.4 (2.3×10^{-5})	0.0 (1.0)
2-link [0.95,1.7](S)	27.2 (7.5×10^{-6})	55.8 (3.4×10^{-7})	-17.9 (9.9×10^{-8})	19.9 (2.4×10^{-5})	11.3 (5.0×10^{-5})

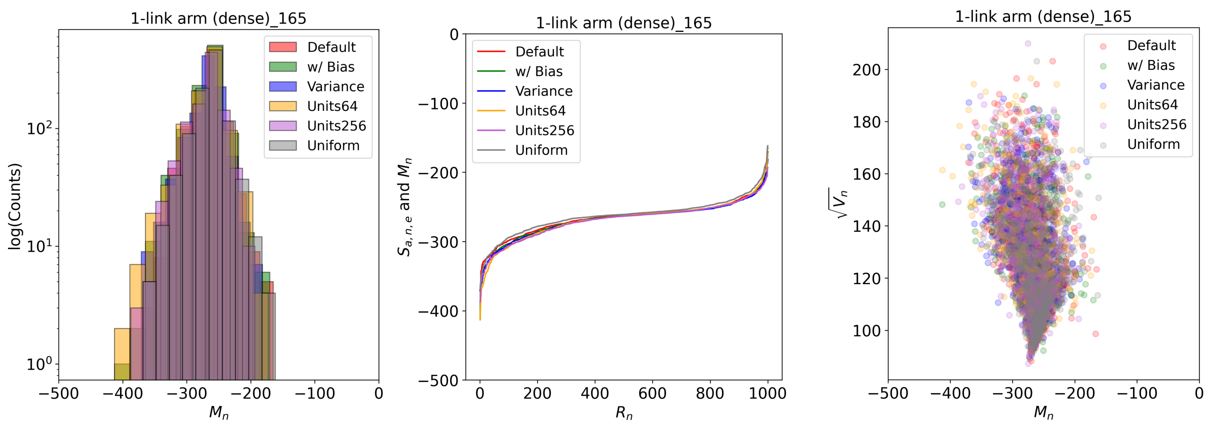

Figure 2: Performance distribution plots for the *1-link* (L = 1.65 m) arm with dense rewards across various policy network architectures and initialisation methods of weights. This shows how the random policies behave similarly across various settings.

4 2-LINK ARM WITH OBSTACLE

We added another task setting, where an obstacle is introduced in the workspace of the *2-link* arm with dense rewards. This enriches our framework with another structurally similar task. We expect that the task with an obstacle should be harder than without the obstacle. Figure 3 depicts the task setting. The red circle is the obstacle with a radius r_o located at position (x_o, y_o). Similar to other tasks, friction and gravity are ignored, and we assume all links have a thickness of t_o. The aim of the task is for the end-effector (arm-tip) to reach an arbitrary target (within the workspace) while avoiding collision with the obstacle.

We can define the distance between the each link and the obstacle as:

$$\begin{aligned} d_i &= \min_{p \in link_i} \operatorname{dist}(p, O) \\ &= \min_{p \in link_i} \left[\|p - (x_o, y_o)\| - (r_o + 0.5 t_o) \right] \end{aligned} \tag{7}$$

where O is the obstacle region defined by radius r_o and p is any point lying along the i^{th} link. Starting from the ground (a) to the end-effector (c), the first link has $p = a + \frac{k}{N}(b - a)$, while the second link has $p = b + \frac{k}{N}(c - b)$. Note that a, b, c represent coordinates of each joint and the end-effector. N are the number of sample points along a link with index $k \in [0, N]$. Our definition of p follows a uniform discretisation of a line segment. When $d_i \leq 0$ (is set to zero if negative) the link and obstacle are in collision, while for $d_i > 0$ the link is collision free. In our setting, $N = 50$, $r_o = 0.02$ m, $t_o = 0.05$ m, $L_1 = 0.95$ m and $L_2 = 0.7$ m.

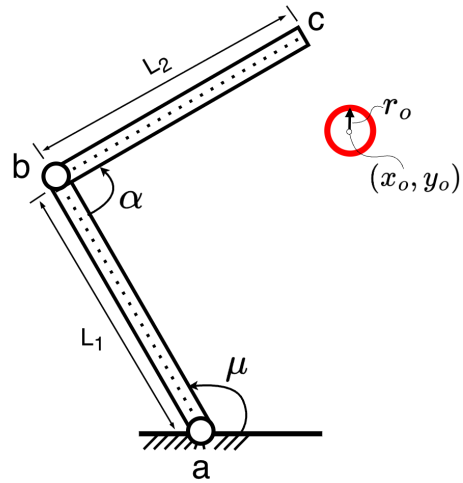

Figure 3: *2-link* arm with obstacle in the workspace.

The reward function that best captures the task aim is given by:

$$r = -\omega_1 \|P_{ee} - P_g\|_2^2 - \omega_2 \|action\|_2^2 - \beta_1 \mathbb{I}_{collision} - \beta_2 \sum_{i=1}^{2} e^{-\alpha d_i} \tag{8}$$

where $[\omega_1, \omega_2] = [1, 1]$ are distance and control weights. P_{ee} and P_g are respectively end-effector and target/goal positions. $[\beta_1, \beta_2] = [10^3, 5]$ are collision and proximity weights and $\alpha = 2.5$ is a smoothing factor. The collision penalty is

$$\mathbb{I}_{collision} = \begin{cases} 1, & \text{if any link collides with obstacle} \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

while the proximity penalty is expressed as the sum of exponentials $\sum_{i=1}^{2} e^{-\alpha d_i}$. During implementation, d_i is restricted from taking values below zero, such that $e^{-\alpha d_i} \in [0, 1]$. This means $e^{-\alpha d_i}$ approaches unity when $d_i \longrightarrow 0$ and approaches zero when $d_i \gg 0$. Note that the obstacle position was constant for the entire training run and the environment episodes terminated under the following three conditions:

(1) When collision occurs, i.e. $d_i = 0$.
(2) When the end-effector is within threshold distance from the target, i.e. $\|P_{ee} - P_g\|_2 \leq 0.05$ m.
(3) When a maximum of 50 steps is reached.

The learning curves of the task along with others is presented in Figure 4. We can see from the plots that SAC takes longer to converge and converges at return values below the rest, especially *2-link* arm with dense-rewards and no obstacle. This shows that the task is harder than its counterpart.

In Figure 5, the *mean performance histogram* is right-skewed with the mode (i.e. most frequent value) being less than 50%. The *mean performance curve* plot depicts two bands on episodic returns, one below 0.4 and another above 0.6. When collision with the obstacle did not occur, policies managed to reach peak performance in some episodes thus behaving similar to a *2-link* arm (dense-rewards) without obstacles. However, when collision occurred, performance was limited to 40% of the maximum return. This underscores the effect of an obstacle in the environment. The *variance distribution* illustrates how the highest variance was experienced at approximately the centre of the range of the mean performance, and the lowest variance is below the centre. This shows that when the obstacle was encountered ($M_n < 0.5$), performance was bounded in comparison to collision-free episodes. Figure 5 is consistent with Figure 4 in showing that this task is harder than the *2-link* arm with dense-rewards and without any obstacles.

By studying Table 4, we notice that the PIC value of the *2-link* arm task with an obstacle is the lowest amongst tasks in dense-rewards settings. This meets our expectations, since the metric implies that the task is the hardest amongst others with dense rewards. However, the POIC value of the task is the highest amongst its counterparts. This denotes that finding the optimal behaviour in this task is easier than all other tasks. This conflicts with results in Figure 4, thus showing another instance of unreliable task hardness ranking by the metric. Results of statistical significance using the Welch's t-test (Delacre et al., 2017) are presented in Table 5. This shows strong statistical significance since the p-values of the t-statistic are in the orders of 10^{-5} below a typical cut-off p-value of 0.005 (Benjamin et al., 2018).

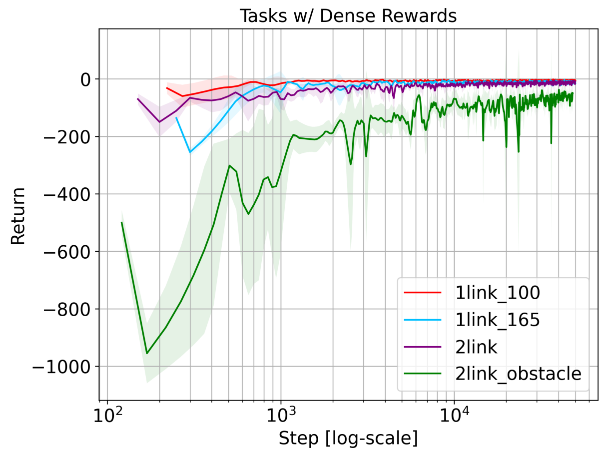

Figure 4: Learning curves of SAC algorithm across tasks with dense rewards. To accommodate wide and varying ranges of steps, results are plotted on a logarithmic scale to enhance interpretability. The results are obtained via evaluation of each task over 5 runs. Note that the plots seem to have different starting steps due to varying sampling rates.

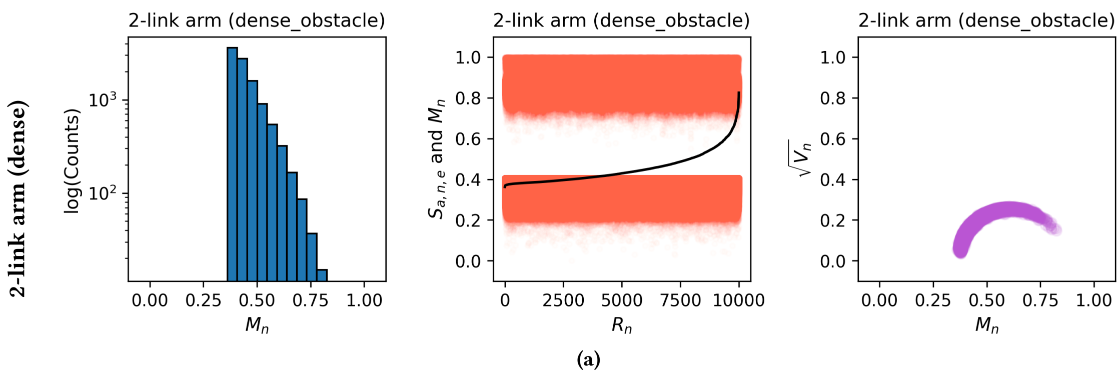

Figure 5: Performance distribution plots for the *2-link* arm with dense rewards and an obstacle in the workspace. The left column shows a histogram of mean performances of the random policies (*Log-scale histogram of* M_n). The middle column depicts *mean performance curves* in black, i.e. mean performance M_n vs rank R_n. Moreover, all the cumulative rewards of the policies $s_{a,n,e}$ across the trials are represented by red dots (behind the black curve). The right column displays plots of standard deviation $\sqrt{V_n}$ vs mean performance M_n (often referred to as *variance distribution*). The plots were made using 10^4 random policies.

Table 4: PIC and POIC values with $N = 10^4$ samples (random policies). High PIC and POIC values correspond to easier tasks, while low values correspond to harder tasks.

Rewards	Arm [dim]	PIC ($\times 10^{-3}$)	POIC ($\times 10^{-3}$)
Dense	1-link [1.0]	4005 ± 8.5	2.628 ± 0.056
	1-link [1.65]	4153 ± 8.4	4.105 ± 0.085
	2-link [0.95,1.7]	4200 ± 6.1	0.725 ± 0.011
	2-link [obstacle]	3671 ± 14.3	21.31 ± 0.33
Sparse	1-link [1.0]	85.11 ± 0.4	1.958 ± 0.034
	1-link [1.65]	71.21 ± 0.2	1.197 ± 0.031
	2-link [0.95,1.7]	45.95 ± 0.0	0.946 ± 0.0079

Table 5: Statistical significance between PIC and POIC values using Welch's t-test. Results are presented as: *statistic (p-value).* This are for only task with dense-rewards.

	1-link [1.0]	*1-link [1.65]*	*2-link [0.95,1.7]*	*2-link [obstacle]*
		PIC		
1-link [1.0]	0.0 (1.0)	-	-	-
1-*link [1.65]*	-20.1 (2.3×10^{-7})	0.0 (1.0)	-	-
2-link [0.95,1.7]	-29.3 (2.9×10^{-8})	-8.6 (2.6×10^{-5})	0.0 (1.0)	-
2-link [obstacle]	37.4 (5.4×10^{-9})	48.5 (3.8×10^{-11})	55.4 (1.3×10^{-11})	0.0 (1.0)
		POIC		
1-link [1.0]	0.0 (1.0)	-	-	-
1-link [1.65]	-17.7 (1.2×10^{-7})	0.0 (1.0)	-	-
2-link [0.95,1.7]	30.8 (4.5×10^{-6})	59.7 (2.6×10^{-7})	0.0 (1.0)	-
2-link [obstacle]	-85.1 (1.3×10^{-8})	79.0 (2.6×10^{-8})	-97.5 (6.3×10^{-8})	0.0 (1.0)

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